

Longitudinal changes in cortical grey matter in adolescents with developmental language disorder

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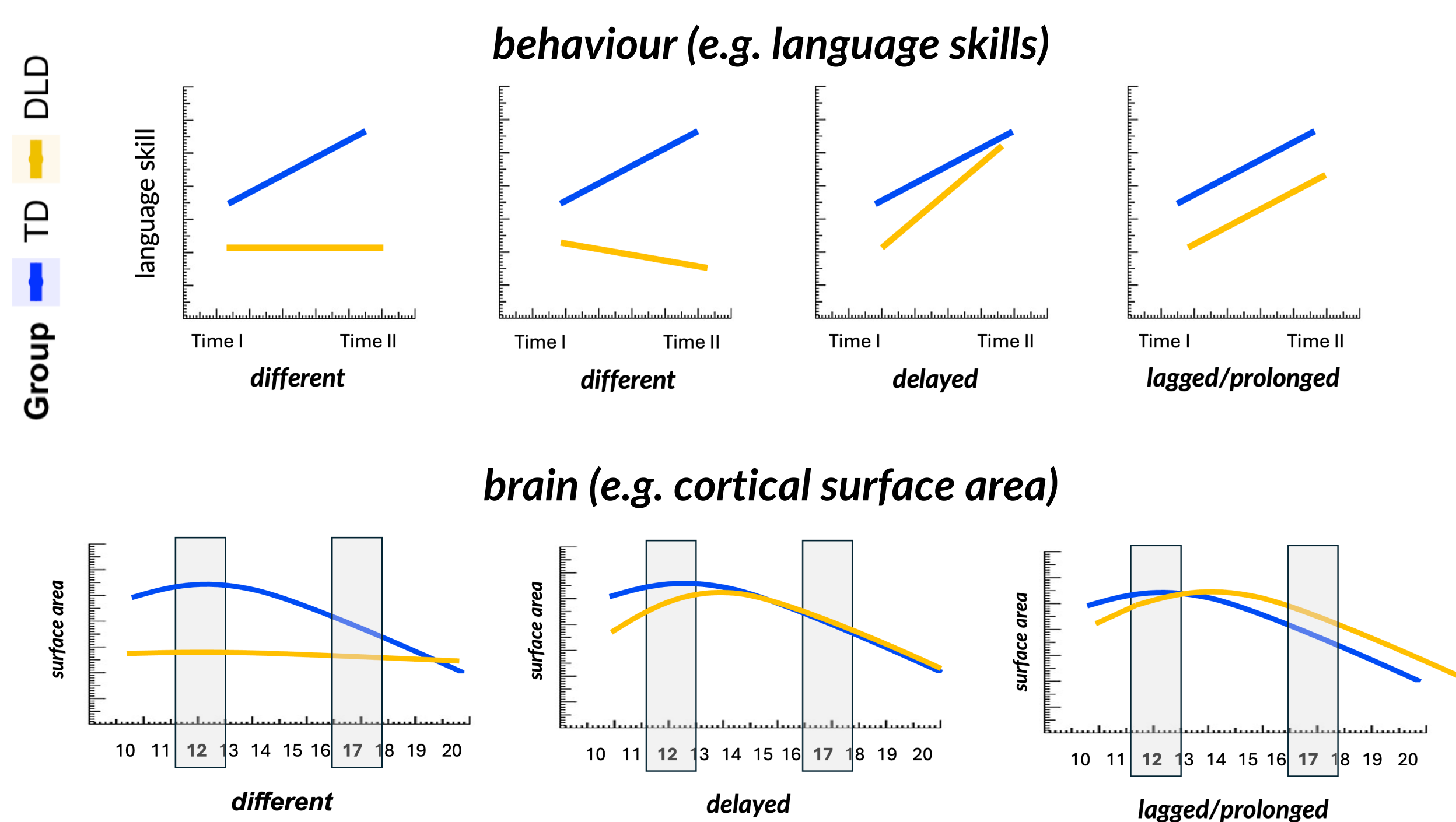
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Background & Research Goals

- Developmental language disorder (DLD) is characterised by persistent difficulties in learning language.
- Previous cross-sectional research describes a behavioural profile of difficulties across a range of different language tasks, with considerable heterogeneity across individuals. Neuroimaging studies report atypical grey matter morphology in peri-sylvian regions that are critical for speech and language.
- Longitudinal research is needed, particularly in adolescence and adulthood.

- Investigate trajectories of change in behaviour and whether language difficulties persist, plateau, or continue to develop over time.
- Investigate how cortical thickness and surface area evolve over development in DLD.

- We tested three possible trajectories of change in DLD:

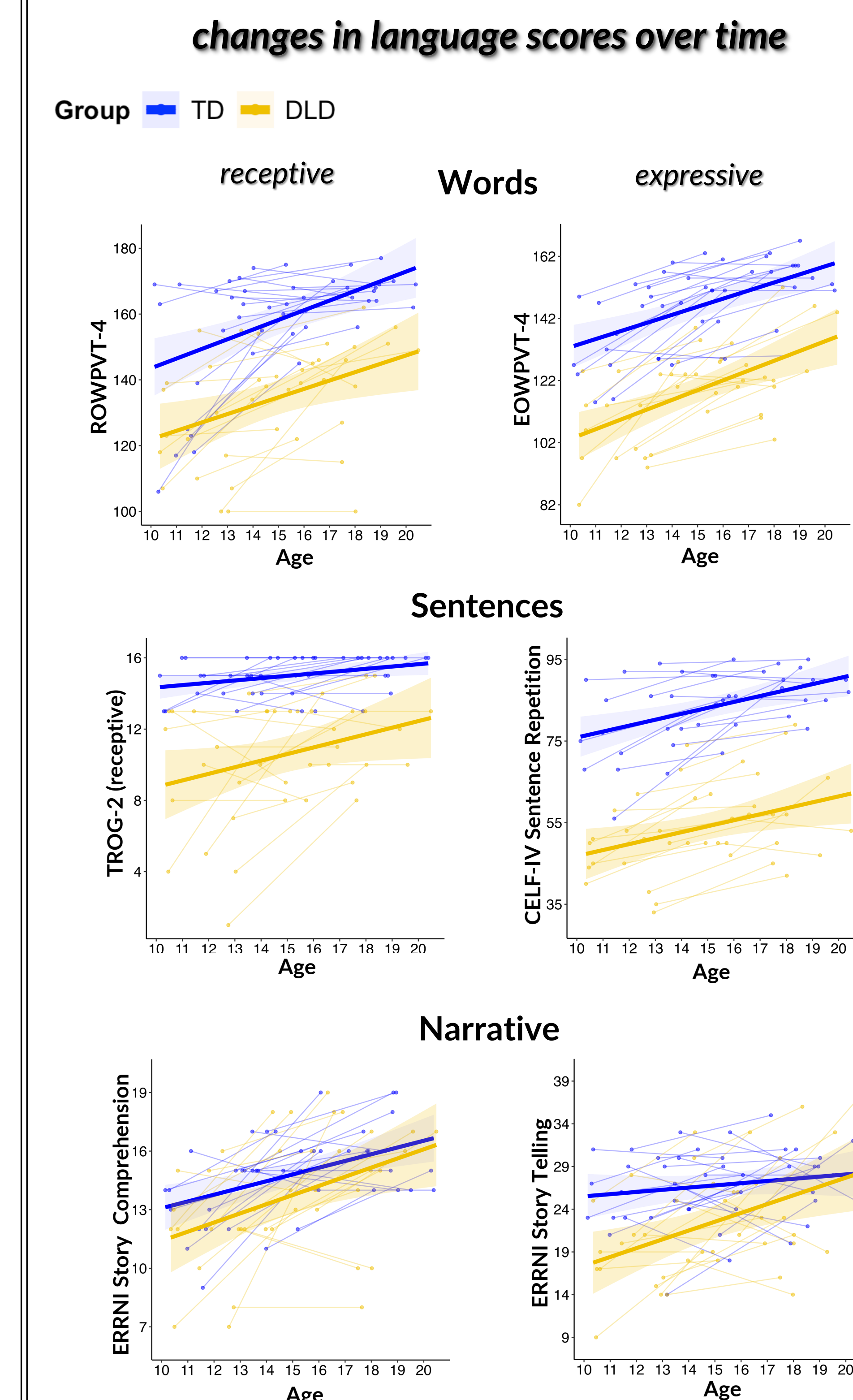


Participants & Group Definitions

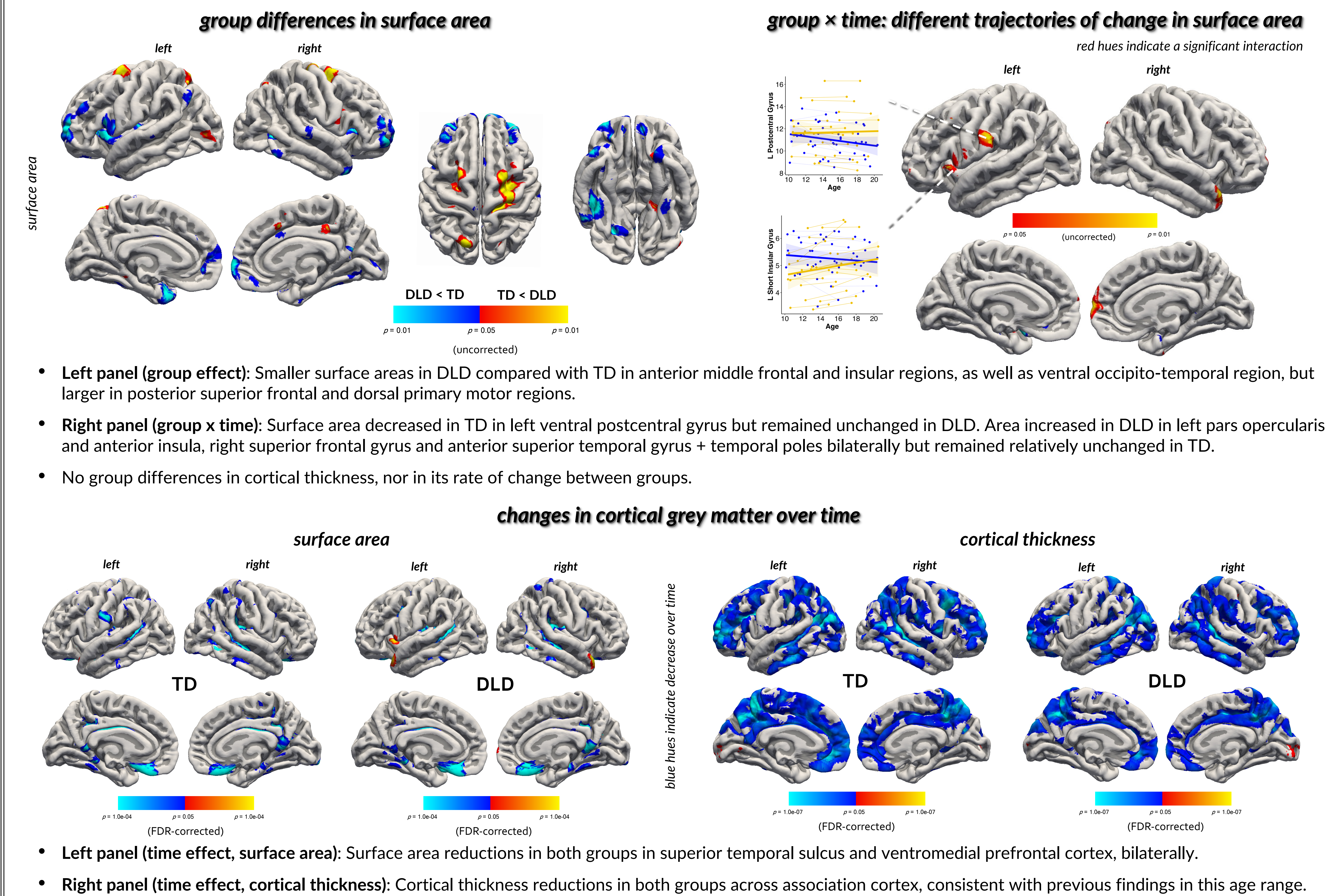
- N = 53 Native English-speaking school-age children.
- Age at Time I: 10-15 years ($m = 12.61$ y).
- Age at Time II: 14-20 years ($m = 17.22$ y).
- Mean interval between sessions = 4y6m (range: 3y9m – 5y11m).

	Nonverbal reasoning >70	2+ language tests failed	History of speech & language problems
22 TD 11 Male 18 R-handed	✓	✗	✗
20 DLD 13 Male 14 R-handed	✓	✓	✓

Results



- Similar rates of progress in both TD and DLD in language scores over time, but a consistent performance gap remained in DLD in tests of vocabulary and grammar, especially on sentence repetition test (proxy for expressive grammar).



- Left panel (group effect):** Smaller surface areas in DLD compared with TD in anterior middle frontal and insular regions, as well as ventral occipito-temporal region, but larger in posterior superior frontal and dorsal primary motor regions.
- Right panel (group x time):** Surface area decreased in TD in left ventral postcentral gyrus but remained unchanged in DLD. Area increased in DLD in left pars opercularis and anterior insula, right superior frontal gyrus and anterior superior temporal gyrus + temporal poles bilaterally but remained relatively unchanged in TD.
- No group differences in cortical thickness, nor in its rate of change between groups.

Materials & Methods

Neuropsychological Measures

- Expressive and receptive vocabulary, grammar, and narrative skills.

Statistical Analyses

- Behavioural:** Linear mixed models for raw scores; predictors = group, time, group $\times$ time; covariates = age, sex; random intercepts = participant.
- Neuroimaging:** Spatiotemporal linear mixed models (*fslmer* package); metrics = cortical thickness, surface area, grey matter volume; predictors & covariates as above + eTIV.

MRI Acquisition & Image Processing

- Prisma 3T, 3D T1w MP-RAGE.
- FreeSurfer v7.4.0 (longitudinal pipeline); subject-specific templates; 10 mm FWHM smoothing.
- QC: Visual inspection + Qoala-T at each stage.

Scan for references
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Discussion

- Behaviour:** Qualitatively similar developmental patterns in DLD group similar to TD controls, but at consistently lower levels across most language skills, reflecting a persistent performance lag extending into adolescence and young adulthood.
- Brain:** Greater surface area expansion in DLD group in left pars opercularis and anterior insula may reflect experience-dependent plasticity and late functional specialization.
- Longitudinal data spanning a wider age range and incorporating multiple time points will be required to characterise cortical trajectories in this population.